



Dynamics of cell aggregates fusion: Experiments and simulations



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HIGHLIGHTS

- Experiments of cell aggregates fusion are performed using human adipose derived stem cells.
- The process of fusion is simulated using the cellular Potts model with aggregates reaching up to 10,000 cells each.
- The simulations imply that individual cell motion corresponds to an anomalous diffusion.
- Cell diffusion has a strong influence on the dynamics of aggregates fusion.

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ABSTRACT

Fusion of cell tissues is an ubiquitous phenomenon and has important technological applications including tissue biofabrication. In this work we present experimental results of aggregates fusion using adipose derived stem cells (ADSC) and a three dimensional computer simulation of the process using the cellular Potts model with aggregates reaching 10,000 cells. We consider fusion of round aggregates and monitor the dimensionless neck area of contact between the two aggregates to characterize the process, as done for the coalescence of liquid droplets and polymers. Both experiments and simulations show that the evolution of this quantity obeys a power law in time. We also study quantitatively individual cell motion with the simulation and it corresponds to an anomalous diffusion.

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1. Introduction

The study of fusion of cell aggregates is important in biofabrication and its *in silico* modeling can help the design of bioengineered tissues. Fusion is also important in different developmental processes including heart formation [1,2].

Aggregates of cells or multicellular spheroids behave similarly to liquids in long time scales: they round to minimize their surface and fuse or envelop each other when placed in contact [3,4]. In these types of experiments the spheroids are grown in culture medium containing all required cell nutrients. They have been used to model tumors [5,6] and more recently used as 'building blocks' for organ tissue engineering [7]. In this case, a tissue is fabricated from several spheroids that are entrapped in hydrogel, forming a three dimensional matrix. The expected result would be the fusion of the aggregates forming a uniform tissue. Hence, a detailed description of cell aggregates fusion is a necessary first step to yield the adequate control

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necessary for biofabrication. To acquire control of these processes we investigate the physics subjacent to the observed cell behavior. Cells in an aggregate interact in complex ways that involve (i) adhesion molecules that may differentially attach to different cell types, (ii) active motion caused by the cytoskeleton activity and (iii) the cortical tension originating from the cytoskeleton structure that causes cells resist to deformations. In terms of rheological properties aggregates present simultaneously elastic, viscous and plastic properties [8]. For example, aggregates generated by random mixtures of two or more cell types undergo a sorting process where the final configuration is one layer of a tissue enveloping the other (engulfment), resembling the phase separation in pure physical systems due to interfacial free energy minimization [9,10]. Forgacs and coworkers have studied fusion using a simplified simulation where cells are represented as dots, which is an oversimplification of a real cell [11]. More recently, Rieu and coworkers (see supporting information in Ref. [8]) experimentally determined the speed with which the contact area increases during the fusion of two cell aggregates.

Today, there is a growing experimental and theoretical consensus that a balance between cell adhesion and cortical tension, which determines the shapes of cells in epithelia, can play a relevant role in the dynamics of cell aggregates and tissue organization [9,12–14]. In this work we present new experimental data on fusion and we study with numerical simulations the influence on the process of several mechanisms including cell adhesion, motility, cortical tension and the aggregate number of cells.

2. Methods

2.1. Experiments

Cell culture: adipose tissue derived stem cells (ADSC) from Invitrogen were used for this experiment. Cells were multiplied in MesenPro RS basal medium (Invitrogen) supplemented with 2% MesenPro RS growth supplement (Invitrogen), 1% Penicillin/Streptomycin (Invitrogen), 1% L-glutamine (Invitrogen), 1% Amphotericin B (Sigma-Aldrich). Cells were maintained in a humidified incubator at 37 °C in 5% CO₂–95% air atmosphere, with the medium changed every second day. Cells from first passage were used in experiments.

Tissue spheroid production: 32 tissue spheroids were produced in parallel by the traditional hanging drop method. Cells from confluent ADSC layers were harvested using 0.25% Trypsin/EDTA for 5 min in a humidified incubator. They were counted and the concentration of cell suspension was readjusted to 10⁶ cell/ml (25,000 cell/25 μl). From this suspension droplets of 25ml were placed on the lid of Petri dishes that were inverted and incubated at 37 °C in humidified incubator for 48h. During this period of time, tissue spheroids formed at the bottom of the droplets due to gravity. Then the tissue spheroids were juxtaposed in hanging drops with fresh medium barely touching each other. The fusion of tissue spheroids was followed up for 2 days and pictures were taken with a Dage-MTI RC300 camera mounted on a Leica MZ12 stereomicroscope at every 4 h.

2.2. Numerical simulation

The simulation model was developed using the CompuCell3D 3.6 simulator [15] of the cellular Potts (CPM) which is available for download from: <http://www.compuCell3d.org>

The model is defined as follows. A label is attributed to each site of a discrete lattice with dimensions starting from $L_x \times L_y \times L_z = 63 \times 63 \times 106$ to $L_x \times L_y \times L_z = 74 \times 74 \times 312$ voxels. Cells or external medium are represented by domains composed by neighboring labels with the same value. The energy between cell–cell or cell–medium interactions is proportional to the shared interface length with energy $E_{\tau\tau'}$ per unit area, where τ specifies cell type and/or medium. The energy (in arbitrary units) used in the Monte Carlo process over the cubic lattice is defined as

$$E = \frac{1}{2} \sum_{\langle ijk, i'j'k' \rangle} E_{\tau_{ijk}\tau'_{i'j'k'}} \times I_{\tau_{ijk}\tau'_{i'j'k'}} + \lambda_V \sum_{\tau} (V(\tau) - V_T)^2 + \lambda_S \sum_{\tau} (S(\tau) - S_T)^2, \quad (1)$$

where $E_{\tau_{ijk}\tau'_{i'j'k'}}$ is the interfacial energy per unit area between neighboring lattice sites labeled by τ_{ijk} and $\tau'_{i'j'k'}$ and $I_{\tau_{ijk}\tau'_{i'j'k'}}$ is the area of interface between them that depends on the range of interaction among lattice sites. The sum in the first term extends to third-nearest neighbors that decreases significantly lattice energy anisotropy. The last energy terms constrains the volume V and surface S [12,13,16] of each cell to predefined target values $V_T = 125$ voxels and $S_T = 190$ pixels, with intensity controlled by λ_V and λ_S , respectively. The last term in Eq. (1) models the cell cortical tension. External medium is represented by an unconstrained generalized cell occupying all the empty space. The initial state of the simulation is two spherical aggregates of identical cells touching each other (see Fig. 1). The system evolves using a Monte Carlo process with a Metropolis dynamics explained as follows. A site σ of the lattice and one of its 6 nearest neighbors are randomly chosen to test local boundary energy minimization. If the two sites are of different types, then we propose that the site assumes the neighboring label value σ' and accept it with probability P determined by the Metropolis algorithm: a variation in energy ΔE is accepted with probability 1, if $\Delta E \leq 0$. If $\Delta E > 0$ a second random number, r , is generated and the change of label occurs if $e^{-\Delta E/T} > r$, otherwise the label is not changed. T is a fluctuation parameter that controls cell motility and is kept constant during the simulation. Higher T implies higher motility and vice versa. This process is repeated indefinitely. The unit of time, one Monte Carlo step (MCS), is defined as the number of exchange attempts equal to the total number of sites of the lattice. Only cell

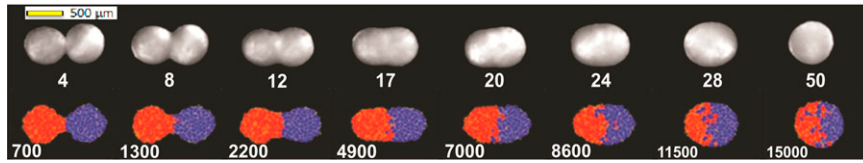


Fig. 1. Top: snapshots of a fusing pair of two-day-old cell aggregates of ADSC. Aggregates fusion and complete rounding up takes approximately 2 days and the aggregates an estimated number of cells close to 20,000. Snapshots taken at 4, 8, 12, 17, 20, 24, 28 and 50 h. Bottom: cross section snapshots of the fusion simulation at 700, 1300, 2200, 4900, 7000, 8600, 11,500, 15,000 MCS. Aggregates have 361 cells each. Colors are used to distinguish cells belonging to different aggregates in simulations.

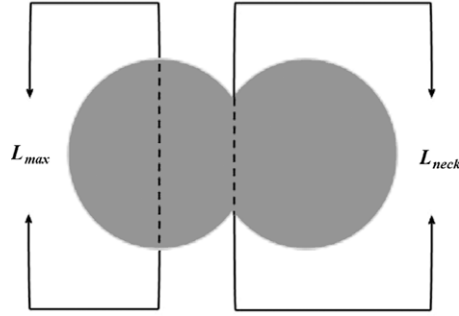


Fig. 2. After they were put into contact, the neck L_{neck} and the maximum L_{max} lengths depicted between the two aggregates were measured at 4, 8, 12, 17, 20, 24, 28 and 50 h and the squared ratio $(L_n/L_{max})^2$ was taken for each of the 36 different experiments performed.

interaction energies contribute to the total energy in Eq. (1). Their values are $E_{cc} = 2$ and $E_{cm} = 6$ for cell–cell and cell–medium interactions, respectively. E_{mm} is not relevant since the medium does not contact itself. When not mentioned, the results were generated with aggregates having 361 cells each. In this case 1 h corresponds to approximately 150 MCS. Simulations were performed on Linux personal computers with Intel i7 processors and lasted approximately between 4 and 40 h.

3. Results

Fig. 1 compares one fusion experiment with numerical simulations. In the upper part a sequence of photos taken during 50 h shows some steps of the fusion process of two aggregates of Adipose Derived Stem Cells (ADSC, see Section 2). The lower part shows the numerical simulation equivalent sequence during 15,000 MCSs. A movie of this simulation in Fig. 1 is available as [Supplementary material](#) to this article. A total of 32 aggregates were produced in parallel for the experiment. The visualization of every one is similar to that presented in Fig. 1. The comparison between simulations and experiments were made as follows. The experimental images were analyzed and the contact length between the aggregates $L_{neck}(t)$ and their maximum diameter $L_{max}(t)$ were measured as a function of time. Fig. 2 illustrates how these quantities were measured. We then took the squared ratio between them. From our simulations, we obtained the instantaneous contacting planar area between the two aggregates $A_{neck}(t)$ and divided it by its instantaneous maximal area $A_{max}(t)$. The comparison was made between these two ratios and it is shown in a log–log plot in Fig. 3, for different values of the motility parameter T . We are interested in comparing the curve behaviors before saturation to extract the exponent values. The crossing of the curves in Fig. 3 and others is the sole result of the effect of the normalization used to allow a comparison of the lines. The analysis indicates that both experimental and simulated data obey a power law. The exponent obtained from a linear fitting to the experimental data is ~ 1.2 and the error bars are about the size of the symbols on graph. From Fig. 3 we can also see that the exponent of the simulated curves is mainly regulated by the cell diffusion controlled by parameter T . λ_V and λ_S are model parameters that constrain the energy associated to cell volume V and surface S to predefined target values (see Section 2.2). λ_S affects the fusion process and this is shown in Fig. 4. For a given λ_V , when λ_S increases, initially the exponent changes strongly and then has a slow increase. To simulate the experimental exponent using $\lambda_S = 0$ requires very high T and in this case we find that the fusion process is dominated by mixing of the cells from the different aggregates due to high cellular motility in disagreement with experiments [11]. According to Ref. [11] cell mixing is negligible until the aggregates coalesce forming the pattern corresponding to 20 h in Fig. 1.

For the study of the influence of the cell motility parameter and the constraints on cell volume and cell surface we generated a cubic tissue with periodic boundary conditions. The cubes have dimensions of 55^3 and 1300 cells, 85^3 and 5000 cells, 105^3 and 10,000 cells. Particle diffusion obeys Eq. (2), the diffusion constant D and the diffusion exponent α are related by

$$D = \langle \Delta r^2 \rangle / (6t^\alpha), \quad (2)$$

where t is time and $\langle \Delta r^2 \rangle$ is the mean square displacement (MSD) of the cells. α is a positive exponent that characterizes correlations in the motion. In Table 1 we list the exponent α obtained from Eq. (2). For $\alpha > 1$ the motion is classified as

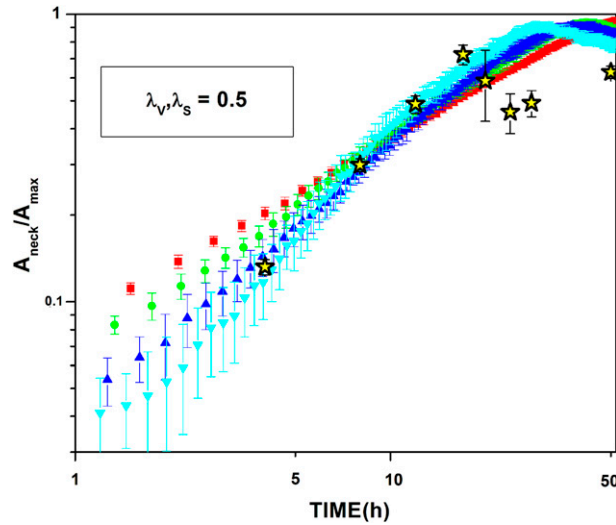


Fig. 3. Log-log plot of the contact area between the aggregates $A_{\text{neck}}/A_{\text{max}}$ as a function of time for different values of the fluctuation parameter and for an equal value (0.5) of volume and surface constraints. The square of the contact length $(L_{\text{neck}}/L_{\text{max}})^2$ between the aggregates is plotted to match dimensionally the area ratio obtained from simulations. The experimental data are superimposed on simulation data for sake of slope comparison. Stars (★) represent experimental data from two-day-old aggregates of adipose derived stem cells. Simulated aggregates have 361 cells each. Values of the fluctuation parameter used are 20 (■), 45 (●), 70 (▲) and 90 (▼).

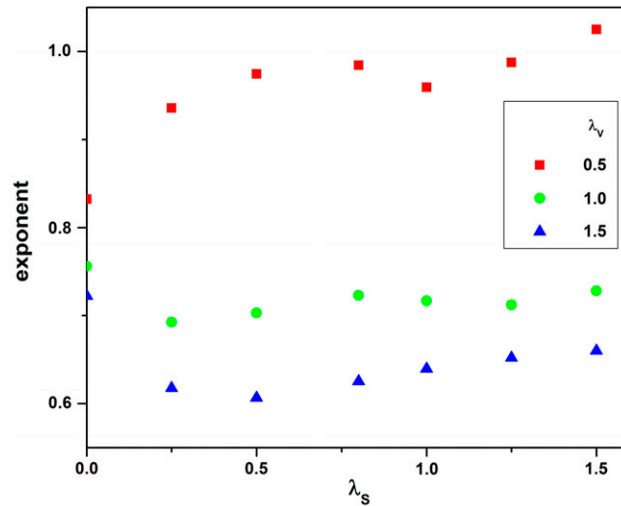


Fig. 4. Plot of the exponents of the simulation for the fusion curve for different values of the constraints on volume and cell surface. Simulated aggregates have 361 cells each. $\lambda_v = 0.5$ (■), 1 (●) and 1.5 (▲). $T = 70$.

superdiffusive, for $\alpha < 1$ the motion is subdiffusive and for $\alpha = 1$ the motion is uncorrelated. The results show that cells apparently have normal diffusive behavior ($\alpha \sim 1$) only when cortical tension is absent ($\lambda_s = 0$) [17]. In all other cases the motion is subdiffusive, with α influenced by the cortical tension. The diffusion results are independent of the number of cells and this is shown in Fig. 5. Evidence of correlated motion was obtained by Glazier and coworkers in *Hydra* cells [18,19] where they found superdiffusive behavior. The authors explained the origin of this correlated diffusion by the fact that cells do not move independently since they are in close contact with each other. In Fig. 6 we show the effect of the constraints on the diffusion constant of the cells in a semi-log plot. The diffusion constant D correlates strongly with the cortical tension (λ_s) decreasing exponentially with it. Remarkably, for very low values of D we observe that fusion still proceeds to completion with very low cell mixing. We conclude that both the cortical tension and T have influence on cell motion in the simulation and on the amount of mixing in fusion. An influence of the cortical constraint on cell motion should be expected as the cellular membrane elasticity is affected and changes the packing density of the cells in the aggregate [19].

The effect of the number of cells of the aggregate on the fusion exponent is shown in Fig. 7. The exponent decreases slowly for larger aggregates tending to constancy, see Fig. 9 and Table 2. The change of the exponent with aggregate number of cells is probably a finite size effect of the simulations.

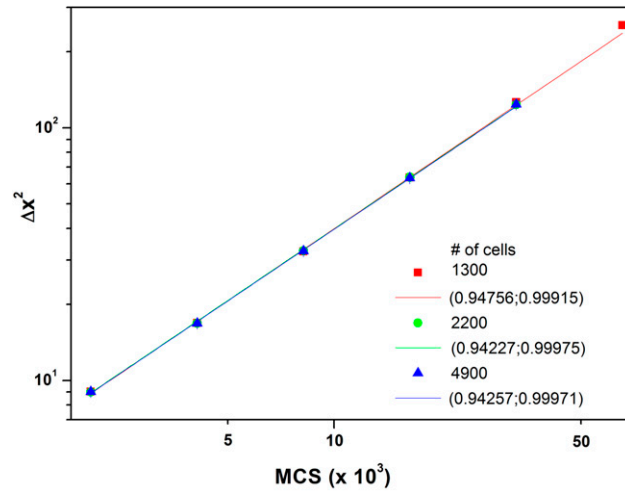


Fig. 5. Plot of the MSDs for aggregates sizes of 1300 cells (■), 2200 cells (●), and 4900 cells (▲) each. Corresponding colored lines stand for linear fittings. In the parenthesis we present the slope and the correlation coefficient for each fitting. Simulation parameters: $\lambda_V = \lambda_S = 0.5$, $T = 70$.

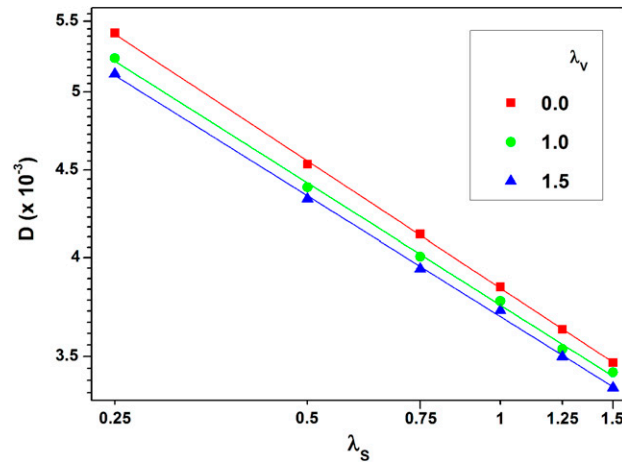


Fig. 6. Semi-log plot of the diffusion constant of the cells in the simulation for the fusion curve for different values of the constraints on volume and cell surface. Simulated aggregates have 361 cells each. $\lambda_V = 0.5$ (■), 1.0 (●) and 1.5 (▲). $T = 70$.

Table 1
Diffusion exponent α .

λ_V	λ_S	0	0.25	0.5	0.75	1.0	1.25	1.5
0.5		1.04 ± 0.01	0.832 ± 0.009	0.77 ± 0.01	0.73 ± 0.02	0.72 ± 0.02	0.71 ± 0.01	0.71 ± 0.02
1.0		1.04 ± 0.01	0.835 ± 0.009	0.76 ± 0.01	0.73 ± 0.02	0.71 ± 0.02	0.71 ± 0.01	0.70 ± 0.02
1.5		1.04 ± 0.01	0.837 ± 0.009	0.76 ± 0.01	0.73 ± 0.02	0.71 ± 0.02	0.70 ± 0.02	0.70 ± 0.02

Effect of the constraints on volume and cell surface (Eq. (1)) on the simulation fusion exponent α (Eq. (2)). Averages calculated over 15 simulations.

The effect on the fusion curve of the increase of adhesion between cells is shown in Fig. 8. The curves (and error bars) overlap significantly suggesting that changes in cell adhesion have a much weaker effect on the exponent when compared to cell motility.

4. Discussion

In this paper we presented experimental results on the dynamics of cell aggregates fusion using ADSC cells and the CPM to simulate the process. Experiments and simulations show power law behavior in the dynamics of tissues fusion. Through simulations we studied how the biological mechanisms affect the exponent of the power law. We predict from the simulations that the exponent is affected by cell motility, seems to converge to a constant value with increasing number of

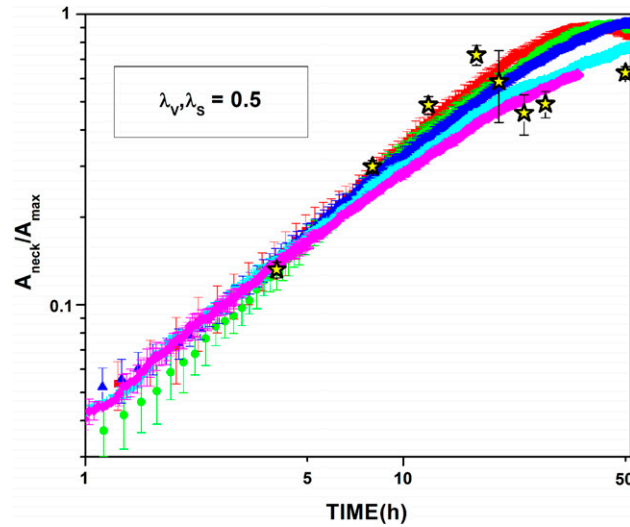


Fig. 7. Log–log plot of the ratio of the contact area between the aggregates and its maximal area $A_{\text{neck}}/A_{\text{max}}$ as a function of time for different sizes of the aggregate and for an equal value (0.5) of volume and surface constraints. The square of the contact length $(L_{\text{neck}}/L_{\text{max}})^2$ between the aggregates is plotted to match dimensionally the area ratio obtained from simulations and experimental data ($\star$). The experimental data is superimposed on simulation data for sake of exponent comparison. Aggregate sizes are 361 cells ($\blacksquare$), 831 cells ($\bullet$), 1096 cells ($\blacktriangle$), 5000 cells ($\blacktriangledown$), 10,000 cells ($\blacklozenge$). $T = 70$.

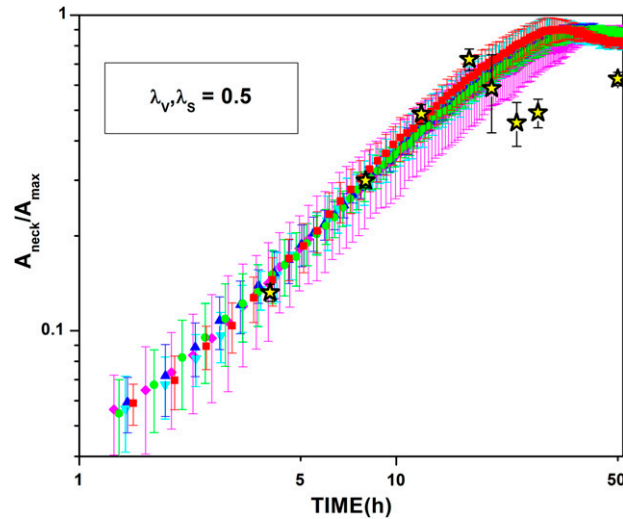


Fig. 8. Log–log plot of the ratio of the contact area between the aggregates and its maximal area $A_{\text{neck}}/A_{\text{max}}$ as a function of time for different values of cell–cell adhesion energies (E_{cc}) and for an equal value (0.5) of volume and surface constraints. The square of the contact length $(L_{\text{neck}}/L_{\text{max}})^2$ between the aggregates is plotted to match dimensionally the area ratio obtained from simulations and experimental data ($\star$). The experimental data is superimposed on simulation data for sake of exponent comparison. Simulated aggregates have 361 cells each. ($\blacksquare$) $E_{\text{cc}} = 1$, ($\bullet$) $E_{\text{cc}} = 1.5$, ($\blacktriangle$) $E_{\text{cc}} = 2$, ($\blacktriangledown$) $E_{\text{cc}} = 3$, ($\blacklozenge$) $E_{\text{cc}} = 3.5$. $T = 70$.

Table 2

Fusion exponent vs. aggregate size.

Size	Exponent
361	0.972 ± 0.002
831	0.947 ± 0.002
1096	0.931 ± 0.002
5000	0.870 ± 0.002
10000	0.843 ± 0.002

Fusion exponent for the curves in Fig. 7.

Averages calculated over 10 simulations.

cells of the aggregate and is less affected by changes in cell adhesion. The increase of T increases the amount of mixing of cells belonging to the different aggregates speeding up the process. To our knowledge, no experiments have yet investigated

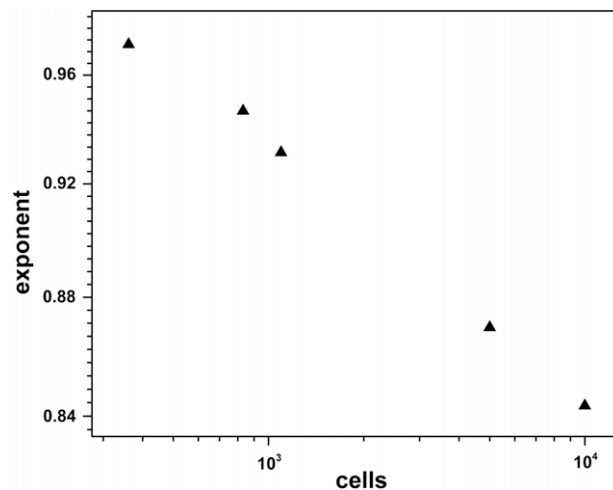


Fig. 9. Plot of the fusion exponents in Fig. 7 vs. aggregate size. Averages calculated over 10 simulations.

this result, however the model predicts that fusion can proceed to completion even if cells have almost passive membranes – that is, membrane ruffling is absent (see Fig. 5). Some experiments have observed cell motion in fusion [11] and they reported that cell mixing does not play a major role in the process – however a detailed study is still missing. The fact that fusion could proceed to completion with passive cell membranes is an interesting prediction of our simulation and remains to be investigated experimentally. We believe that it could be tested using the drug cytochalasin as performed in some studies of cell sorting [14,20].

In the field of polymer sintering where the fusion of two polymer drops (analogous to aggregates fusion) is studied for technological applications, the Frenkel's model is highly quoted, see Ref. [21,22]. According to this 2D model the radius x of the contacting neck between two fusing drops of polymers evolves as a power law in time as

$$x/a \sim \left(\frac{\gamma t}{\eta a_0} \right)^{0.5},$$

where γ is the polymer surface tension, η viscosity, a is the drop radius, a_0 is the initial drop radius and t is the time. In this model the exponent is constant and the process is faster with higher surface tension and retarded by viscosity and drop size. Despite the intrinsic differences in nature between the two systems, our simulation results do not seem to obey a similar law. The results show an exponent affected mainly by cell motility and weakly by cell adhesion. A recent study compared Frenkel's model with cell aggregates but the quantitative agreement is not completely satisfactory (see support information in Ref. [8]). A possible explanation for the origin of the discrepancy between Frenkel's model and the CPM is that there is no true viscosity in the Potts model and this could explain the origin of the disagreement.

From the analysis of cell motion, we showed that cells perform normal diffusion only when the cortical tension term is null or maybe very weak, otherwise it is subdiffusive (Table 1). We believe this change from normal to anomalous diffusion arises from a change in the packing density of cells in the aggregates mainly influenced by the cortical constraint. Studies of cell motion in *Hydra* aggregates [18,19] confirm correlations, however they are superdiffusive and we are not able to explain the origin of this discrepancy at the moment. Recent 2D simulations detected only normal diffusion in cell motion [17] but apparently for lower values of the cortical constraint when compared to values used in this study.

We believe that our results bring useful information to the field of tissue engineering since it characterizes how different cellular mechanisms can influence the fusion of tissues and their artificial design.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.physa.2013.10.037>.

References

- [1] J.M. Perez-Pomares, R.A. Foty, Tissue fusion and cell sorting in embryonic development and disease: biomedical implications, *Bioessays* 28 (2006) 809–821.
- [2] A. Wessels, D. Sedmera, Developmental anatomy of the heart: a tale of mice and man, *Physiol. Genomics* 15 (2003) 165–176.
- [3] R.Z. Lin, Recent advances in three-dimensional multicellular spheroid culture for biomedical research, *Biotechnol. J.* 3 (2008) 1172–1184.

- [4] J.C.M. Mombach, D. Robert, F. Graner, G. Gillet, G.L. Thomas, M. Idiart, J.-P. Rieu, Rounding of aggregates of biological cells: experiments and simulations, *Physica A* 352 (2005) 525–534.
- [5] Y. Jiang, J. Pjesivac-Grbovic, C. Cantrell, J.P. Freyer, A multiscale model for a vascular tumor growth, *Biophys. J.* 89 (2005) 3884–3894. <http://dx.doi.org/10.1529/biophysj.105.060640>.
- [6] A.N. dos Reis, J.C.M. Mombach, M. Walter, L.F. de Avila, The interplay between cell adhesion and environment rigidity in the morphology of tumors, *Physica A* 322 (2003) 546–554.
- [7] K. Jakab, A. Neagu, V. Mironov, R.R. Markwald, G. Forgacs, Engineering biological structures of prescribed shape using self-assembling multicellular systems, *Proc. Natl Acad. Sci. USA* 101 (9) (2004) 2864–2869. <http://dx.doi.org/10.1073/pnas.0400164101>.
- [8] P. Marmottant, A. Mgharbel, J. Käfer, B. Audren, J.-P. Rieu, J.-C. de Viala, B. van der Sandenc, A.F.M. Marée, F. Graner, H. Delanoë-Ayari, The role of fluctuations and stress on the effective viscosity of cell aggregates, *Proc. Natl Acad. Sci. USA* 106 (41) (2009) 17271–17275. <http://dx.doi.org/10.1073/pnas.0902085106>.
- [9] M. Krieg, Y. Arboleda-Estudillo, P.-H. Puech, J. Käfer, F. Graner, D.J. Müller, C.-P. Heisenberg, Tensile forces govern germ-layer organization in zebrafish, *Nature Cell Biol.* 10 (4) (2008) 429–436. <http://dx.doi.org/10.138/ncb1705>.
- [10] J.-D. Muller, M. Bousmina, A. Maazouz, 2d-sintering kinetics of two model fluids as drops, *Macromolecules* 41 (2008) 2096–2103. <http://dx.doi.org/10.1021/ma702349s>.
- [11] E. Flenner, L. Janosi, B. Barz, A. Neagu, G. Forgacs, I. Kosztin, Kinetic Monte Carlo and cellular particle dynamics simulations of multicellular systems, *Phys. Rev. E* 85 (2012) 031907–1–031907–10.
- [12] G.W. Brodland, The differential interfacial tension hypothesis (dith): a comprehensive theory for the self-rearrangement of embryonic cells and tissues, *J. Biomech. Eng.* 124 (2002) 188–197.
- [13] J. Käfer, T. Hayashi, A.F.M. Marée, R.W. Carthew, F. Graner, Cell adhesion and cortex contractility determine cell patterning in the drosophila retina, *Proc. Natl Acad. Sci. USA* 104 (47) (2007) 18549–18554. <http://dx.doi.org/10.1073/pnas.0704235104>.
- [14] M.L. Manning, R.A. Foty, M.S. Steinberg, E.-M. Schoetz, Coaction of intercellular adhesion and cortical tension specifies tissue surface tension, *Proc. Natl Acad. Sci. USA* 107 (28) (2010) 12517–12522. <http://dx.doi.org/10.1073/pnas.1003743107>.
- [15] J.A. Izaguirre, R. Chaturvedi, C. Huang, T. Cickovski, J. Coffland, G. Thomas, G. Forgacs, M. Alber, G. Hentschel, S.A. Newman, J.A. Glazier, CompuCell, a multi-model framework for simulation of morphogenesis, *Bioinformatics* 20 (7) (2004) 1129–1137. <http://dx.doi.org/10.1093/bioinformatics/bth050>.
- [16] N.B. Ouchi, J.A. Glazier, J.P. Rieu, A. Upadhyaya, Y. Sawada, Improving the realism of the cellular potts model in simulations of biological cells, *Physica A* 329 (2003) 451–458.
- [17] A. Nakajima, S. Ishihara, Kinetics of the cellular potts model revisited, *New J. Phys.* 13 (2011) 033035.
- [18] J.-P. Rieu, A. Upadhyaya, J.A. Glazier, N.B. Ouchi, Y. Sawada, Diffusion and deformation of single *hydra* cells in cellular aggregates, *Biophys. J.* 79 (2000) 1903–1914.
- [19] A. Upadhyaya, J.-P. Rieu, J.A. Glazier, Y. Sawada, Anomalous diffusion and non-Gaussian velocity distribution of *hydra* cells in cellular aggregates, *Physica A* 293 (2001) 549–558.
- [20] J.C.M. Mombach, J.A. Glazier, R.C. Raphael, M. Zajac, Quantitative comparisons between differential adhesion models and cell sorting in the presence and absence of fluctuations, *Phys. Rev. Lett.* 75 (1995) 2244–2247.
- [21] J. Frenkel, Viscous flow of crystalline bodies under the action of surface tension, *J. Phys.* 9 (1945) 385–391.
- [22] C.T. Bellehumeur, M. Kontopoulou, J. Vlachopoulos, The role of viscoelasticity in polymer sintering, *Rheol. Acta* 37 (2013) 270–278.